

Seungyeon Back



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Research Interest: LLM Reasoning, LLM Agent, RAG

- 📖 M.S. student in Artificial Intelligence at [Hanyang University](#), advised by Prof. [Dong-Kyu Chae](#), with a B.S. in Mathematics.
Developed a user-adaptive and reliable LLM-based system for personalized dictionaries (Co-first author, *AdapDict*, CSCW 2026) and currently optimizing graph-based retrieval for LG Electronics' real-time recommendations.

Education

🎓 **Hanyang University, Seoul, South Korea (Sep 2025 - present)**

M.S. in Artificial Intelligence

- GPA: 4.25 / 4.5

🎓 **Hanyang University, Seoul, South Korea (Mar 2021 - Aug 2025)**

B.S. in Mathematics

- Minors: Big Data Convergence, Biomedical Engineering
- GPA: 3.97 / 4.5 (Cum Laude)

Research Experience

Co-first Author, AdapDict: A Level- and Domain-Adaptive Educational Dictionary and Encyclopedia System (CSCW 2026 Accepted)

- Proposed a level- and domain-adaptive explanation framework for controllable explanation generation in LLMs.
- Designed a generator-verifier loop with structured outputs, reducing hallucination by 50%.
- Developed a user-adaptive prompting mechanism for multi-dimensional control of explanation difficulty.
- Conducted a user study (n=41) demonstrating improved clarity, usability, and domain-specific understanding over baseline tools.

Graduate Researcher, GraphRAG-based Recommendation System (in collaboration with LG Electronics) (Jan 2026 – Present)

- Built a real-time data pipeline for news and OTT content ingestion and processing.
- Designed a dual-graph framework aligning news-content graphs via shared entities.
- Developed a graph-based retrieval pipeline using multi-hop traversal to generate candidate content.
- Integrated LLMs for path-based reasoning over graph structures, enabling explainable recommendations.

Co-first Author, An Empirical Study of the Clustering of Mountain Hiking GPS Trajectory Data (Under Revision at IEEE Access)

- Proposed a novel trajectory similarity measure integrating spatial, temporal, and velocity components for irregular spatio-temporal data.
- Designed an ϵ -ball-based spatial similarity metric and a data-adaptive weighting framework (α , β , γ) for flexible similarity modeling.
- Applied hierarchical clustering to large-scale GPS trajectory data to uncover latent behavioral patterns.

Undergraduate Researcher, Applied Statistics Lab, Hanyang University (Nov 2023 – Dec 2024)

- Implemented LSTM, GRU, and BERT-based models for stock prediction using PyTorch, incorporating technical indicators and news data.
- Participated in collaborative study sessions on time-series modeling and machine learning, including peer-led presentations and discussions.

Skills

- **Programming Languages:** Python, R
- **Frameworks:** PyTorch, TensorFlow, HuggingFace Transformers
- **Tools:** Git, Linux
- **Web (Basic):** Flask, HTML, CSS, JavaScript